

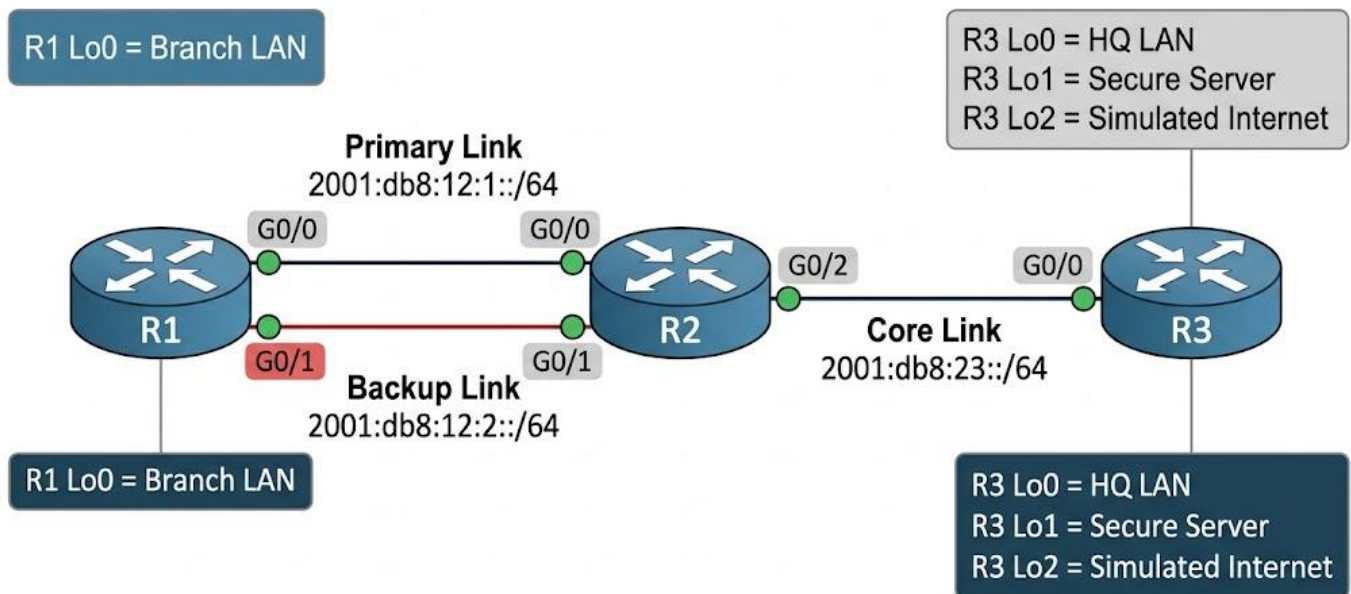
CCNA v2.0 – IPv6 Static Routing & Troubleshooting Lab

Network routes • Host routes • Default route • Floating static route

1. Lab Objectives

- Configure IPv6 addressing on three Cisco routers using loopbacks to simulate LANs and servers.
- Configure and verify an IPv6 network static route (/64).
- Configure and verify an IPv6 host static route (/128).
- Configure a primary IPv6 default route and a floating static default route.
- Build the return path so end-to-end IPv6 connectivity works before and after the primary-link failure.

IPv6 Branch-to-HQ Network Topology



Addressing Table

Router	Interface	IPv6 Address	Purpose
R1	G0/0	2001:db8:12:1::1/64	Primary link
R1	G0/1	2001:db8:12:2::1/64	Backup link
R1	Lo0	2001:db8:10::1/64	Branch LAN
R2	G0/0	2001:db8:12:1::2/64	Primary link
R2	G0/1	2001:db8:12:2::2/64	Backup link
R2	G0/2	2001:db8:23::2/64	Core link
R3	G0/0	2001:db8:23::3/64	Core link
R3	Lo0	2001:db8:30::1/64	HQ LAN
R3	Lo1	2001:db8:30:A::100/128	Secure Server
R3	Lo2	2001:db8:9999::1/64	Simulated Internet

Step 1 – Basic IPv6 Configuration

Configure IPv6 unicast routing and IPv6 addresses on all routers.

R1 Configuration

```
enable
configure terminal
hostname R1
ipv6 unicast-routing

interface g0/0
  description PRIMARY-LINK-T0-R2
  ipv6 address 2001:db8:12:1::1/64
  no shutdown
exit

interface g0/1
  description BACKUP-LINK-T0-R2
  ipv6 address 2001:db8:12:2::1/64
  no shutdown
exit

interface loopback0
  description BRANCH-LAN
  ipv6 address 2001:db8:10::1/64
exit
end
```

R2 Configuration

```
enable
configure terminal
hostname R2
ipv6 unicast-routing

interface g0/0
  description PRIMARY-LINK-T0-R1
  ipv6 address 2001:db8:12:1::2/64
  no shutdown
exit

interface g0/1
  description BACKUP-LINK-T0-R1
  ipv6 address 2001:db8:12:2::2/64
  no shutdown
exit

interface g0/2
  description CORE-LINK-T0-R3
  ipv6 address 2001:db8:23::2/64
  no shutdown
exit
end
```

R3 Configuration

```
enable
configure terminal
hostname R3
ipv6 unicast-routing

interface g0/0
  description CORE-LINK-T0-R2
  ipv6 address 2001:db8:23::3/64
  no shutdown
```

```

exit

interface loopback0
  description HQ-LAN
  ipv6 address 2001:db8:30::1/64
exit

interface loopback1
  description SECURE-SERVER
  ipv6 address 2001:db8:30:A::100/128
exit

interface loopback2
  description SIMULATED-INTERNET
  ipv6 address 2001:db8:9999::1/64
exit
end

```

Step 2 – Verify Directly Connected Networks

```

show ipv6 interface brief
show ipv6 route connected
show ipv6 route

```

Expected: each router should show its own directly connected IPv6 networks. Fix addressing or interface status before continuing.

Step 3 – Network Static Route (/64)

R2 needs a route to the entire Branch LAN behind R1.

```

R2(config)# ipv6 route 2001:db8:10::/64 2001:db8:12:1::1

```

Verify:

```

R2# show ipv6 route static
R2# ping 2001:db8:10::1

```

Step 4 – Host Static Route (/128)

R1 needs a route to only the Secure Server address on R3.

```

R1(config)# ipv6 route 2001:db8:30:A::100/128 2001:db8:12:1::2

```

Verify:

```

R1# show ipv6 route static
R1# show ipv6 route 2001:db8:30:A::100

```

9. Step 5 – Primary and Floating Default Routes on R1

The primary default route uses the R1–R2 primary link. The second route has a higher administrative distance, so it is used only when the primary route is unavailable.

```

R1(config)# ipv6 route ::/0 2001:db8:12:1::2
R1(config)# ipv6 route ::/0 2001:db8:12:2::2 10

```

Verify:

```
R1# show ipv6 route ::/0
R1# show ipv6 route static
```

Expected: the active default route should use 2001:db8:12:1::2 with AD 1.

Step 6 – Return Routes

R2 must have a backup route to R1's Branch LAN so that return traffic still works when the primary R1–R2 link is down.

```
R2(config)# ipv6 route 2001:db8:10::/64 2001:db8:12:2::1 10
```

R3 needs a route toward the R1 Branch LAN. Use a primary route through R2. A second route is not necessary for the R3-to-R1 return path because R2 itself has the primary/backup choices.

```
R3(config)# ipv6 route 2001:db8:10::/64 2001:db8:23::2
```

R3 also needs routes back to the R1-side networks for direct testing. The simplest student-friendly option is a default route toward R2:

```
R3(config)# ipv6 route ::/0 2001:db8:23::2
```

Step 7 – End-to-End Verification

```
R1# ping 2001:db8:30:A::100
R1# ping 2001:db8:30::1
R1# ping 2001:db8:9999::1
```

```
R2# ping 2001:db8:10::1
R2# ping 2001:db8:30::1
```

```
R3# ping 2001:db8:10::1
```

Step 8 – Floating Static Route Failure Test

This is the main CCNA troubleshooting exercise.

```
R1# show ipv6 route ::/0
```

Record the active next-hop and administrative distance.

Now simulate failure of the primary link:

```
R1(config)# interface g0/0
R1(config-if)# shutdown
end
```

Check the routing table:

```
R1# show ipv6 route ::/0
```

Expected: the active default route should change to 2001:db8:12:2::2 with AD 10.

Test reachability:

```
R1# ping 2001:db8:9999::1
R1# ping 2001:db8:30:A::100
```

Restore the primary link:

```
R1(config)# interface g0/0
R1(config-if)# no shutdown
end
```

Troubleshooting Checklist

- Check interface status: show ipv6 interface brief
- Check IPv6 addresses: show ipv6 interface <interface>
- Check connected routes: show ipv6 route connected
- Check static routes: show ipv6 route static
- Check the exact route: show ipv6 route <prefix>

CCNA v2.0 Concepts Covered

Skill	Lab Practice
IPv6 addressing	Global unicast, /64 and /128
Static network route	2001:db8:10::/64
Static host route	2001:db8:30:A::100/128
Default route	::/0
Floating static route	AD 10 backup path
Administrative distance	Primary AD 1 vs backup AD 10
Link-local next-hop	Requires exit interface
Verification	show ipv6 route, ping, interface commands
Troubleshooting	Primary-link failure and route selection

Student Tasks

1. Build the topology in Cisco Packet Tracer.
2. Configure all IPv6 addresses and loopbacks.
3. Configure the network, host, primary default, and floating static routes.
4. Verify end-to-end reachability before failure.
5. Shut down R1 G0/0 and prove that the floating route becomes active.
6. Restore G0/0 and verify that the primary route becomes active again.